

Section Report 4 Summary

Github Link: <https://github.com/aaguilar02/biol4810>

1. FastQC / Falco

The purpose of FastQC is to evaluate the quality of raw sequencing data before downstream analysis is performed. This quality control step allows researchers to identify potential problems within the dataset, including low-quality reads, adapter contamination, GC content abnormalities, sequence duplication, and overrepresented sequences. In this workflow, the Falco tool was used in place of FastQC because it performs the same analysis while processing large datasets more efficiently. The forward and reverse reads both contained approximately 7,728,733 total sequences with a GC content of 63% and a read length of 151 base pairs. No sequences were flagged as poor quality. However, several categories failed quality assessment, including per tile sequence quality, per base sequence content, sequence duplication levels, and adapter content. These results indicated that the dataset required additional trimming and filtering before downstream analysis could be reliably performed.

2. Fastp

The purpose of Fastp is to clean and filter raw sequencing reads in order to improve data quality for downstream analysis. This tool trims low-quality bases, removes adapters, and filters problematic reads that could negatively affect alignment and transcript quantification. Following Fastp processing, the cleaned dataset contained approximately 14.63 Mbp with an average read length of 144 base pairs and a GC content of 63%. Approximately 0.44 million reads were removed during trimming and filtering. The reduction in read number indicates that low-quality or contaminated reads were successfully removed, resulting in a cleaner and more reliable dataset for downstream transcriptomic analysis.

3. Hisat2

The purpose of Hisat2 is to align sequencing reads to a reference genome. This step maps the forward and reverse reads back onto known genomic locations, allowing researchers to determine where transcripts originated within the genome. Aligning reads to the genome is a critical step because it converts raw sequencing data into biologically interpretable information. Accurate alignment allows downstream tools to quantify transcript abundance and identify differences in gene expression between experimental conditions.

4. StringTie

The purpose of StringTie is to assemble transcripts from aligned sequencing reads and estimate transcript abundance. After Hisat2 aligns the reads to the genome, StringTie reconstructs transcripts and determines how strongly each transcript is expressed within each sample. This process converts mapped sequencing reads into transcript-level biological information. By assembling transcripts and estimating expression levels, StringTie provides the foundation necessary for downstream differential expression analysis.

5. FeatureCounts

The purpose of FeatureCounts is to quantify how many sequencing reads map to specific genomic features, such as genes or transcripts. This step converts aligned sequencing reads into count-based gene expression data that can be analyzed statistically. FeatureCounts generates a table of expression counts for each gene across all samples, allowing researchers to compare transcript abundance between experimental conditions. These count tables serve as the primary input for DESeq2 differential expression analysis.

6. DESeq2

The purpose of DESeq2 is to identify genes that are differentially expressed between experimental conditions using RNA-seq count data. Positive log₂ fold-change values indicate upregulated genes, while negative values indicate downregulated genes. The null hypothesis states that there is no difference in gene expression between mutant and wildtype conditions, while the alternative hypothesis states that gene expression differs between the two conditions.

The PCA plot generated from the DESeq2 analysis demonstrated strong separation between mutant and wildtype samples along PC1, which explained 88% of the total variance within the dataset. Biological replicates clustered closely together within each condition, suggesting strong reproducibility and consistency between samples. Similarly, the sample-to-sample distance heatmap showed that mutant replicates clustered together while wildtype replicates formed a separate cluster, further supporting distinct transcriptional differences between conditions.

The dispersion plot estimated variability in gene expression across samples and helped model expression differences throughout the dataset. The histogram of p-values demonstrated that many genes produced low p-values, suggesting that numerous expression differences were statistically significant rather than occurring by random chance. Additionally, the MA plot showed that while many genes remained relatively stable in expression, a subset of genes exhibited strong positive or negative fold changes, indicating substantial upregulation or downregulation between mutant and wildtype samples.

7. Concatenating Multiple Datasets

The purpose of concatenating multiple datasets is to combine separate datasets into a single table for downstream analysis and visualization. This step allows information from different samples or analyses to be merged together into one organized dataset. Combining datasets simplifies downstream processing and allows multiple sources of information to be analyzed simultaneously.

8. Volcano Plot

The volcano plot was used to visualize statistically significant differences in gene expression between mutant and wildtype conditions. Genes with positive log₂ fold-change values represent upregulated genes, while genes with negative log₂ fold-change values represent downregulated genes. The volcano plot demonstrated a substantial number of significantly downregulated genes, along with a smaller group of significantly upregulated genes. Many genes displayed low p-values, indicating that these observed expression differences are unlikely to be due to random variation. The broader spread of downregulated genes suggests that the mutant condition suppresses expression across multiple genes, while a smaller subset of genes becomes more highly expressed.

9. Heatmap

The heatmap was generated to visualize patterns of gene expression across mutant and wildtype samples using color intensity to represent relative expression levels. Genes with similar expression patterns clustered together, while mutant and wildtype samples formed distinct groupings based on their transcriptional behavior. The clear separation between experimental conditions indicates that the mutant and wildtype samples exhibit substantially different gene expression profiles. Additionally, the close clustering of biological replicates further supports the reliability and consistency of the sequencing results.

10. Top 10 Upregulated and Downregulated Genes

The Top 10 Up and Down table was generated to identify genes with the largest changes in expression between mutant and wildtype conditions. Genes with large positive log₂ fold-change values were considered strongly upregulated, while genes with large negative log₂ fold-change values were considered strongly downregulated. The most strongly upregulated genes likely contribute to biological pathways that become activated under the mutant condition, while the most strongly downregulated genes likely represent pathways that are suppressed. Changes in gene expression can contribute to altered phenotypes by affecting important cellular and metabolic processes.

11. Filtering Data Using Simple Expressions

The purpose of filtering data using simple expressions is to isolate specific rows of interest from a larger dataset based on defined criteria. In this workflow, filtering was used to remove entries from the GFF annotation file that were not coding sequences. By selecting only rows that met the desired criteria, the analysis focused specifically on biologically relevant coding regions. This step improved downstream annotation and interpretation by reducing unnecessary or unrelated information.

12. Overall Interpretation

Overall, the RNA-seq workflow successfully identified substantial differences in gene expression between mutant and wildtype conditions. Quality control and trimming improved sequencing reliability, while alignment and transcript assembly converted raw sequencing reads into biologically meaningful information. Differential expression analysis demonstrated clear transcriptional differences between conditions, supported by strong clustering patterns in the PCA plot and heatmap. The volcano plot and MA plot further demonstrated that numerous genes were significantly upregulated or downregulated in response to the mutation. Together, these results indicate that the mutant condition strongly influences global transcriptional activity and alters expression patterns across multiple genes and biological pathways.